Synthesis S. Dagorne Short Review

Recent Developments on *N*-Heterocyclic Carbene Supported Zinc Complexes: Synthesis and Use in Catalysis

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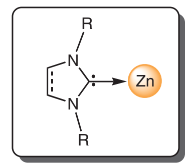

NHC-Zn species

- *Cheap, low-toxicity metal source*
- *Ready synthesis*
- *Steric/electronic tunability*
- *Robust species for various catalyses*

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Abstract The present contribution reviews the synthesis, reactivity, and use in catalysis of NHC–Zn complexes reported since 2013. NHC-stabilized Zn(II) species typically display enhanced stability relative to common organozinc species (such as Zn dialkyls), a feature of interest for the mediation of various chemical processes and the stabilization of reactive Zn-based species. Their use in catalysis is essentially dominated by reduction reactions of various unsaturated small molecules (including CO_2), thus primarily involving Zn–H and Zn–alkyl derivatives as catalysts. Simple NHC adducts of Zn(II) dihalides also appear as effective catalysts for the reduction amination of CO_2 and borylation of alkyl/aryl halides. Stable and well-defined Zn alkoxides have also been prepared and behave as effective catalysts in the polymerization of cyclic esters/carbonates for the production of well-defined biodegradable materials. Overall, the attractive features of NHC-based Zn(II) species include ready access, a reasonable stability/reactivity balance, and steric/electronic tunability (through the NHC source), which should promote their further development.

Key words *N*-heterocyclic carbenes, zinc, catalysis, hydrosilylation, carbon dioxide, reduction, polymerization

1 Introduction

Due to their exceptional σ-donating properties and steric tunability,[1,2] *N*-heterocyclic carbenes (NHC) are ubiquitous ligands for the stabilization of various metal/heteroatom centers across the periodic table, ranging from carbophilic to oxophilic metal/heteroatom derivatives.[3–6] Due to their enhanced stability and the specific steric/electronic properties upon NHC coordination, NHC-supported metal species have found a wide array of applications, ranging from their use in homogeneous catalysis to the design of well-defined organometallics for optical/magnetic applications.[7,8] NHC coordination may also allow the stabilization of unprecedented structural motifs and highly reactive organometallics of potential interest for small-molecule activation.

Organozinc compounds are long-known and widely used reagents for various stoichiometric/catalytic organic transformations, primarily as Lewis acids and reduction/functionalization agents of unsaturated substrates.[9] Although the first NHC–Zn complexes {of the type $[(NHC)ZnEt_2]$} were reported by Arduengo 25 years ago,[10] NHC–Zn compounds and their possible applications have remained relatively unexplored until 5–10 years ago. Two primary factors currently drive the development of NHC–Zn species: (a) the improved stability of such species allows access to robust molecularly well-defined/low-aggregated Zn(II) species (including reactive Zn-based organometallics) of potential interest for various effective/selective transformations, and (b) since zinc is an inexpensive and low-toxic metal source, NHC–Zn(II) catalysts are particularly attractive as alternatives to toxic and/or expensive metal sources.

An earlier review covered NHC-bearing group 12 metal (Zn, Cd, Hg) organometallics up to 2012.[11] Over the past five years (since 2013), the synthesis of structurally diversified

NHC–Zn species and their use in catalysis have attracted increased attention. Such Zn species were mainly exploited in reduction catalysis of unsaturated substrates (ketones, imines, alkenes, alkynes), including challenging substrates such as CO_2, and in polymerization catalysis of cyclic polar monomers. The present contribution reviews the synthesis and use in catalysis of NHC–Zn complexes reported since 2013.

2 NHC-Supported Zinc Alkyl/Aryl Species

2.1 Synthesis

NHC adducts of commercially available ZnR_2 (R = Me, Et) are easier-to-handle ZnR_2 sources, which, given the importance of zinc alkyls as alkylating/reducing reagents, should be of potential interest for various organic reactions. Such adducts are typically prepared by reaction of free NHC with ZnR_2.[10] The resulting Zn–alkyl compounds are either monomeric {[(NHC)ZnR_2]} or dimeric {[(NHC)ZnR_2]$_2$; through two Zn–R–Zn bridges, R = alkyl}, depending on the steric hindrance at the Zn(II) center. All known aryl derivatives [(NHC)Zn(Ar)$_2$] are monomeric. Over the past 4–5 years, a number of monomeric NHC-bearing Zn dialkyls or diaryls (**1** and **2**; Figure 1) have been reported and primarily used as polymerization catalysts (*vide infra*).[12–15] Notably, related mixed alkyl chloro derivatives **3** (Figure 1) may also be prepared in good yield *via* a direct protonolysis/alkane elimination route between the corresponding imidazolium chloride and ZnR_2, thus avoiding the pre-generation of free NHC for adduct formation.[16,17] Ring-expanded NHC–Zn adducts **4** (Figure 2) were also prepared and characterized, as well as a couple of Zn(II) complexes stabilized by NHC-containing C,N,N- and C,N,P-type tridentate ligands (**5** and **6**; Figure 2).[18–20]

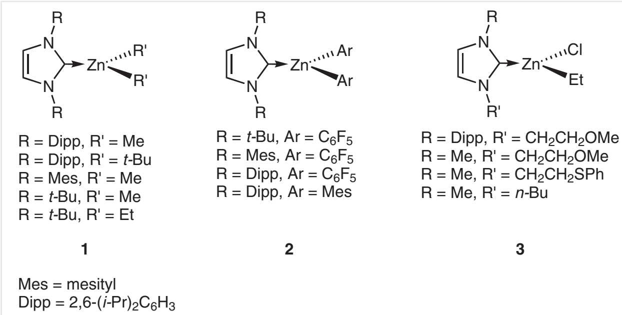


Figure 1 NHC-stabilized Zn dialkyls **1**, diaryls **2**, and mixed alkyl halide species **3**

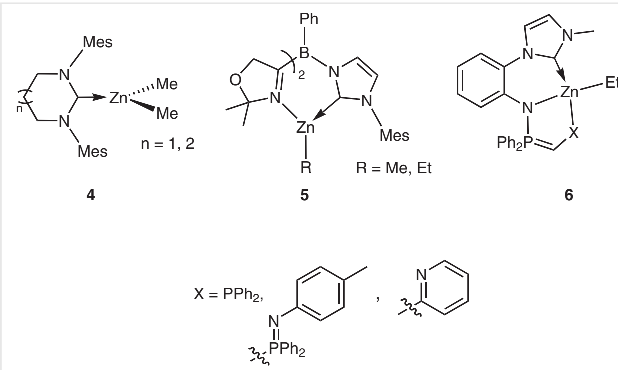
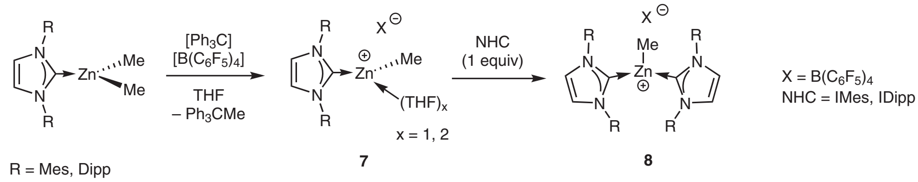


Figure 2 Ring-expanded NHC complexes **4** and C,N,N- and C,N,P-supported Zn species **5** and **6**

NHC coordination was recently shown to impart substantial stabilization to cationic Zn–alkyl species, a class of compounds of interest for the activation of polar/unsaturated substrates due to the enhanced Lewis acidity/electrophilicity at Zn(II). In 2013, the first report of the preparation of NHC-supported Zn–Me cations appeared; this occurred through a Me^- abstraction reaction between a strong Lewis acid {[B(C_6F_5)$_3$] or [Ph_3C][B(C_6F_5)$_4$]} and the corresponding [(NHC)$ZnMe_2$] neutral precursors in the presence of THF, to afford cations [(IMes)ZnMe(THF)$_2$]$^+$ and

Biographical Sketch

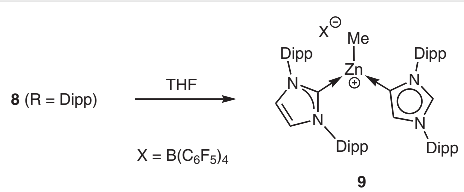
Samuel Dagorne obtained his undergraduate degree at the University of Rennes (France) in 1994. In 1995, he joined the group of Professor Richard F. Jordan at the University of Iowa (Iowa City, USA) and graduated with a Ph.D. in organometallic chemistry in 1999. He then joined Professor Richard R. Schrock's group (MIT, USA) as a postdoc. Back in France, he joined the C.N.R.S. in 2000 and is currently a senior researcher/team leader at the Institute of Chemistry of Strasbourg (University of Strasbourg, France). His research interests include the synthesis, reactivity, and use in catalysis of oxophilic organometallic complexes.



Scheme 1 Synthesis of Zn cations $[(NHC)ZnMe(THF)_x]^+$ (**7**) and $[(NHC)_2Zn–Me]^+$ (**8**)

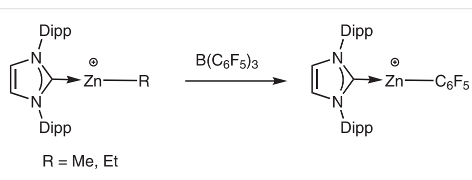
$[(IDipp)ZnMe(THF)]^+$ (**7**; Scheme 1).[12] The addition of NHC (1 equiv) to **7** leads to the formation of the bis-NHC cationic adduct $[(NHC)_2Zn–Me]^+$ (**8**).[21] Interestingly, contrasting with the typical hydrolytic instability of Zn–alkyl cations, the IMes bis-adduct **8** is an air-stable compound in the solid state, reflecting substantial Zn(II) stabilization by the two IMes ligands.

Unlike $[(IMes)_2Zn–Me]^+$, the severely sterically congested $[(IDipp)_2Zn–Me]^+$ is unstable and may readily undergo, for steric relief, a normal- to abnormal-NHC rearrangement to yield cation $[(IDipp)(aIDipp)Zn–Me]^+$ (**9**, *a*IDipp = C4/C5-bonded IDipp; Scheme 2).[21]



Scheme 2 Normal- to abnormal-NHC rearrangement of cation **8** yielding **9**

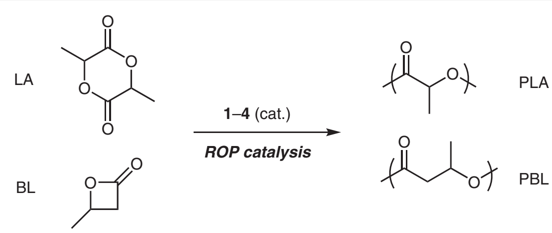
Thanks to NHC stabilization, two-coordinate Zn(II) organocations $[(IDipp)Zn–R]^+$ (**10**, R = Me, Et; Scheme 3) were isolated and characterized from the stoichiometric reaction of $[(IDipp)ZnR_2]$ with $[Ph_3C][B(C_6F_5)_4]$.[22] The more Lewis acidic $[(IDipp)Zn–C_6F_5]^+$ cation (**11**; Scheme 3) may also be accessed *via* an R/C_6F_5 ligand-exchange reaction between $(IDipp)Zn–R^+$ and $B(C_6F_5)_3$. Species **10** and **11** stand as potent Lewis acids and constitute the first two-coordinate Zn(II) organocations to be thus far characterized. Experimental and theoretical data for cation $[(IDipp)Zn–C_6F_5]^+$ suggest a stronger Lewis acidity than that of $B(C_6F_5)_3$.



Scheme 3 Two-coordinate Zn organocations **10** and **11**

2.2 Reactivity and Use in Catalysis

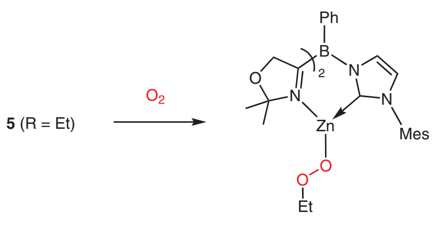
In terms of reactivity, neutral NHC–Zn alkyls **1–4** were essentially used as ring-opening polymerization (ROP) catalysts of cyclic esters such as lactide (LA) and β-butyrolactone (BL) for the production of the derived biodegradable polyesters polylactide (PLA) and poly(β-butyrolactone (PBL), respectively (Scheme 4).[12,17,18]



Scheme 4 Ring-opening polymerization (ROP) of cyclic esters LA and BL catalyzed by adducts **1–4**

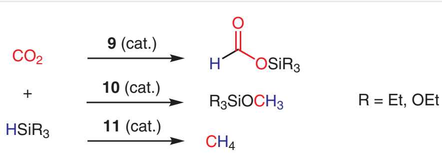
When compared to the average performance of other metal-based catalysts, ROPs mediated by **1–4** typically proceed with moderate activity and chain-length control of the produced polymers {typical run: $[(IDipp)Zn(C_6F_5)_2]$ (1 mol%), LA (100 equiv), r.t., 20 h, conversion to PLA: quant, M_n = 9200 g·mol^{-1}, D = 1.96}. The polymerization process is likely initiated through dissociation of the NHC–Zn bond leading to a Zn-coordinated/activated monomer and free NHC. The latter may then undergo a nucleophilic substitution at the C=O activated cyclic ester, allowing subsequent monomer ring-opening. In the presence of an alcohol source such as BnOH, the Zn–Et chelates **6** (Figure 2) were shown to mediate the controlled ROP of ε-caprolactone (CL) and LA, presumably *via* a coordination–insertion mechanism involving an *in situ* formed Zn–OBn moiety (reaction of **6** with BnOH).[19] Regarding fundamental reactivity, it is also noteworthy that NHC-bearing bis(oxazolyl) Zn–Et species **5** (Figure 2) readily inserts O_2 to yield a rare example of a well-characterized Zn(II) alkylperoxide species **12** (Scheme 5).[20]

Cations **9–11** (Schemes 2 and 3) were found to be effective catalysts for a number of transformations, including CO_2, alkene, and alkyne hydrosilylation catalysis. Tricoordinate cation **9** mediates CO_2 hydrosilylation catalysis in the

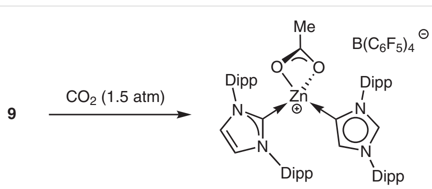


Scheme 5 Formation of NHC–Zn alkylperoxide species **12**

presence of $HSi(OEt)_3$ under low CO_2 pressure (1.5 bar) to yield the mono-reduced formate product $HCO_2Si(OEt)_3$ with moderate activity [**9** (7 mol%), CO_2 (1.5 bar), 90 °C, 60 h, TOF = 0.23 h^{-1}].[21] On the basis of various control experiments, the catalysis is regarded to likely proceed through initial CO_2 insertion to yield the Zn–OAc cation **12** (Scheme 7), which then reacts with $HSi(OEt)_3$ to produce a Zn–H cation, the actual CO_2 hydrosilylation catalyst.



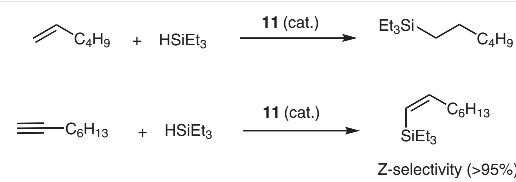
Scheme 6 CO_2 hydrosilylation catalyzed by NHC-supported Zn–alkyl/aryl cations **9–11**



Scheme 7 CO_2 insertion in the Zn–Me$^+$ bond of **9** to afford the Zn–OAc cation **12**

Due to their enhanced Lewis acidity, two-coordinate cations **10** and **11** are much more effective CO_2 reduction catalysts than **9** and display a different product selectivity (Scheme 6). Thus, species **10** selectively hydrosilylates CO_2 in the presence of $HSiEt_3$ to the tris-reduced methanol-equivalent product $MeOSiEt_3$ [**10** (5 mol%), CO_2 (1.5 bar), 90 °C, 12 h, TOF = 1.7 h^{-1}].[22] With the more Lewis acidic cation **11** (5 mol%), CO_2 is completely converted into the tetra-reduced product CH_4 within 6 hours at 90 °C (TOF = 3.4 h^{-1}; Scheme 6). It is a reflection on the stability of such NHC-stabilized Zn(II) cations that a tenfold decrease in the catalyst loading (0.5 mol% of **11**) yielded a 95% conversion into CH_4 within 23 hours (TOF = 8.3 h^{-1}). All experimental and theoretical data on CO_2 hydrosilylation catalysis by catalyst **11** support a Lewis acid type catalysis with the rate-determining step being the addition of $HSiEt_3$ to a Zn-coordinated/activated CO_2 moiety.

Cation **11** catalyzes 1-hexene hydrosilylation at room temperature (100% conversion to $Et_3Si–C_6H_{13}$, 18 h; Scheme 8), while less Lewis acidic Zn–alkyl species **10** is inactive under these conditions.[22] All data hint at a Lewis acid type catalysis, as observed with strong group 13 Lewis acids $[E(C_6F_5)_3]$ (E = B, Al).[23] Cation **11** is also a highly effective catalyst for 1-octyne hydrosilylation, to quantitatively afford the *Z*-selective and mono-reduced silylalkene product [**11** (5 mol%), 15 min, r.t.; Scheme 8], constituting the first instance of Zn-catalyzed alkyne hydrosilylation.[22]



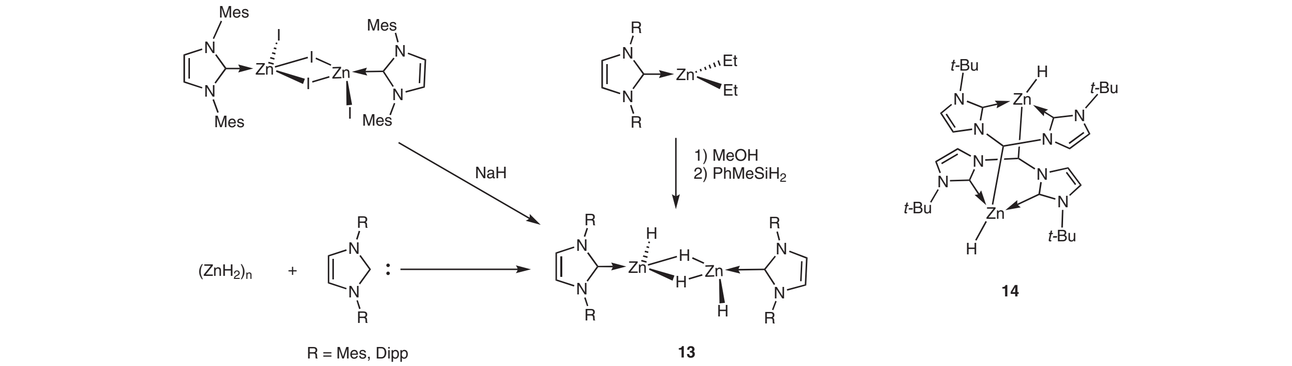
Scheme 8 Alkene/alkyne hydrosilylation catalysis mediated by cation **11**

3 NHC-Supported Zinc Hydride Species

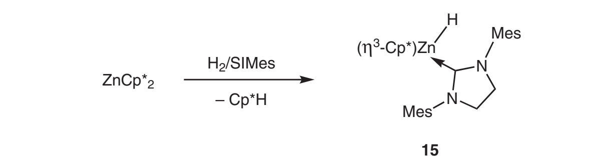
3.1 Synthesis

The low solubility in common organic solvents and limited thermal stability of ZnH_2 prompted the development of ligand-stabilized and molecularly well-defined Zn–H species with improved solubility/stability for use as reducing agents.[24] Discrete adducts of ZnH_2 were only first reported in 2013, upon reacting a stoichiometric amount of IDipp or IMes and ZnH_2 to afford the corresponding $[NHC–ZnH_2]_2$ dimeric adducts **13** (Scheme 9) in good yields.[25] Alternatively, species **13** may be prepared *via* alcoholysis of the Zn–alkyl species NHC–$ZnEt_2$ and subsequent alkoxide/hydride exchange in the presence of $PhMeSiH_2$. Very recently, $[(IMes)ZnH_2]_2$ was also shown to be accessible through a salt metathesis route between the corresponding zinc iodide precursor $[(IMes)ZnI_2]_2$ and NaH (4 equiv).[26]

Dimeric Zn–H species **14**, in which each Zn(II) center is stabilized by a bis-NHC chelating ligand, was synthesized by reaction of the corresponding Zn–Et chelate precursor and Ph_3SiOH and then $PhSiH_3$.[27] Interestingly, access to monomeric NHC-stabilized Zn–H species **15** (Scheme 10) was achieved through hydrogenation of zincocene $[Cp^*{}_2Zn]$ (Cp* = pentamethylcyclopentadienyl) under high H_2 pressure (68 bar) in the presence of the backbone-saturated NHC SIMes.[28] When the reaction was performed with IDipp as an NHC source, the formation of an analogous Zn–H species, i.e., $[ZnCp^*(H)(IDipp)]$, was proposed, though its instability precluded characterization.



Scheme 9 Synthesis of NHC-supported Zn–H species **13** and structure of **14**



Scheme 10 Synthesis of NHC-stabilized Zn–H species **15** through hydrogenolysis

Additionally, several NHC-stabilized mixed alkyl/amido/halo hydrido species were also prepared and structurally characterized following known methodologies for the derivatization of organozinc complexes. They are depicted in Figure 3. For instance, the comproportionation reaction between $[(IMes)ZnH_2]_2$ and $[(IMes)ZnEt_2]$ (1 equiv) led to the formation of the mixed alkyl hydrido $[(IMes)Zn(H)(Et)]_2$ (**16**), while reaction of the zinc dihydride **13** with MeI (2 equiv) afforded the mixed hydrido iodo derivative $[(IMes)Zn(H)(I)]_2$.[29,30] The monomeric amido hydrido species $\{(IDipp)Zn(H)[N(SiMe_3)_2]\}$ (**17**; Figure 3) may be readily produced *via* a hydrido/amido exchange reaction upon mixing stoichiometric amounts of IDipp, the Zn(II) bis-amido precursor $[Zn(N(SiMe_3)_2)_2]$, and $PhSiH_3$ as a hydride source.[31] When $NHMe_2BH_3$ was used instead of $PhSiH_3$, the mixed amido hydrido tetranuclear Zn species **18** (Figure 3) was obtained and structurally characterized.

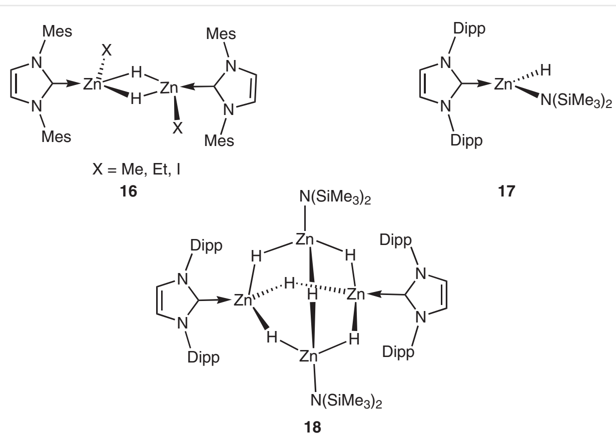
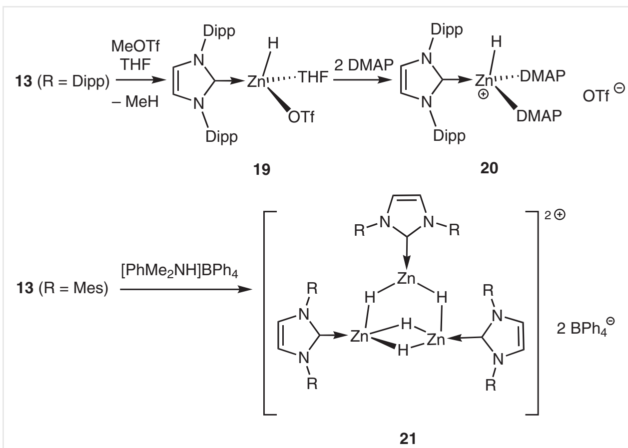


Figure 3 NHC-stabilized mixed alkyl/amido/halo hydrido Zn species **16**–**18**

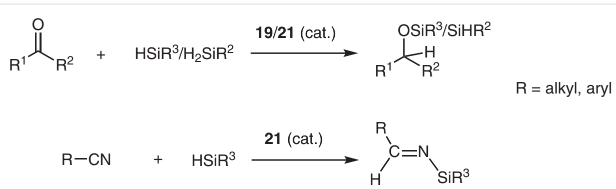
The excellent σ-donating properties of NHCs were further exploited in the stabilization of cationic Zn–H complexes for enhanced electrophilicity/Lewis acidity at Zn(II). In 2014, the NHC-supported Zn–H cation $[(IDipp)Zn(H)(DMAP)_2]^+$ (**20**, DMAP = dimethylaminopyridine; Scheme 11) was reported; it was the first molecularly well-defined Zn–H cation to be isolated and structurally characterized.[30] Cation **20** may be accessed *via* a methane elimination reaction between IDipp-stabilized Zn–H precursor **13** with MeOTf to afford [(IDipp)Zn(H)(THF)(OTf)] (**19**, Scheme 11), which may then be ionized by addition of DMAP (2 equiv) to afford cation **20**. The role of DMAP as a strong Lewis base for the electronic saturation of Zn(II) appears crucial both for the stability and the monomeric nature of cation **20**. In comparison, ionization of $[(IMes)ZnH_2]_2$ with Brønsted acid $[PhMe_2NH]BPh_4$ in the presence of a weaker Lewis base such as THF afforded the dicationic trinuclear Zn–H cluster **21** (Scheme 11) that contains four Zn–H–Zn bridging hydrides.[32]



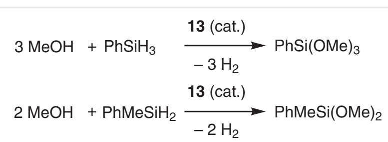
Scheme 11 Synthesis of NHC-stabilized Zn–H cations **20** and **21**

3.2 Reactivity and Use in Catalysis

Reactivity studies of NHC-stabilized molecular Zn–H species with unsaturated substrates showed them to be suitable reagents/catalysts for several transformations, primarily reduction reactions. The reactivity of NHC-bearing Zn(II) hydrides was most studied for carbonyl, imine, nitrile, and CO_2 reduction. Neutral and cationic NHC–Zn hydrides **13**, **14**, **16**, and **19–21** were found to readily and stoichiometrically reduce various carbonyl substrates under mild conditions *via* C=O insertion into the Zn–H bond to afford the corresponding Zn alkoxide species. In addition, cationic Zn hydrides **19** and **21** are highly active in hydrosilylation catalysis at room temperature to produce the corresponding monoreduced silyl ethers at a low catalyst loading [**19** (0.1 mol%), TOF up to 475 h^{-1}, Scheme 12].[30,32] Such high catalytic activity certainly reflects the strongly electrophilic character of the Zn(II) center. Cation **21** (3.3 mol%) also catalyzes the hydrosilylation of nitriles to monoreduced *N*-silylimines within hours at room temperature (Scheme 12), while the neutral Zn–H species **13** mediates the methanolysis of silanes $PhSiH_3$ and $PhMeSiH_2$ to the corresponding silyl ethers along with H_2 generation (Scheme 13).[25,32]



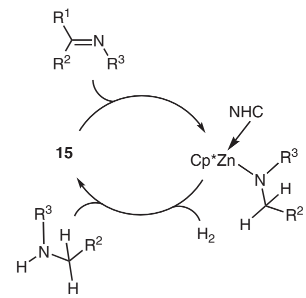
Scheme 12 Catalytic hydrosilylation of ketones and nitriles by cations **19** and **21**



Scheme 13 Catalytic methanolysis of silanes using zinc hydride **13**

The Cp*- and NHC-supported Zn hydride **15**, generated *in situ* from $ZnCp^*_2$ and SIMes (or IDipp), was proposed to be the actual reduction catalyst in Zn-catalyzed imine hydrogenation catalysis (Scheme 14).[28]

More challenging substrates such as CO_2 are readily reduced by NHC–Zn hydrides by CO_2 insertion into the Zn–H bonds. $[(IMes)ZnH_2]_2$ (**13**) and the related mixed methyl hydrido $[(IMes)Zn(H)(Me)]_2$ (**16**) were shown to cleanly afford the corresponding formate insertion products $[(IMes)Zn(O_2CH)_2]_2$ (**22**) and $[(IMes)Zn(O_2CH)(Me)]_2$ (**23**), respectively (Figure 4).[25,29]



Scheme 14 Imine hydrogenation catalysis by Zn species **15**

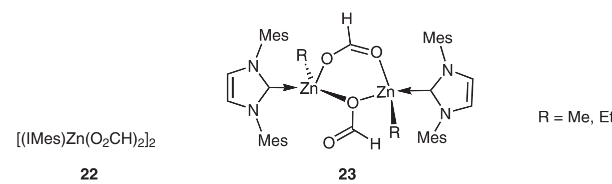
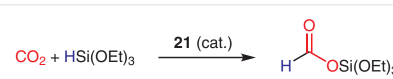


Figure 4 CO_2 insertion products from IMes–Zn hydrides **13** and **16**

In contrast, the reactivity of CO_2 with more sterically congested $[(IDipp)ZnH_2]_2$ is less selective and leads to two products: one arising from CO_2 insertion into the Zn–H bonds, and, the other one, zwitterion [IDipp–CO_2] (possibly due to IDipp dissociation during the reaction).[25] At the catalytic level, trinuclear Zn–H cation **21** was shown to mediate CO_2 hydrosilylation catalysis in the presence of $HSi(OEt)_3$ [**21** (1 mol%), CO_2 (40 bar), 60 °C] with a complete conversion into the monoreduced product $HCO_2Si(OEt)_3$ as the major product (>80%) after 64 hours of reaction (Scheme 15).[32]



Scheme 15 CO_2 hydrosilylation catalysis mediated by Zn–H cation **21**

4 NHC-Supported Zinc Amido/Alkoxide Species

4.1 Synthesis

A few NHC-bearing Zn alkoxides/amides have been characterized over the past five years (Figure 5). Well-defined dimeric benzyloxide species of type **24** were readily prepared in high yields *via* alcoholysis of Zn–alkyl precursors **3**.[16] The dialkoxide adducts $[(IDipp)Zn(OCHR_2)_2]$ (**25**, R = Ph, *t*-Bu) were produced by hydrogenation of $[ZnCp^*_2]$ (4 bar H_2) in the presence of $R_2C{=}O$ and IDipp.[33] Alternatively, protonolysis of $[ZnCp^*_2]$ with R_2CHOH (R = Ph, *t*-Bu; 2 equiv) and subsequent addition of IDipp also allowed the isolation of derivatives **25**. Adducts $\{(NHC)Zn[N(SiMe_3)_2]_2\}$ (**26**) were prepared upon combining the corresponding free NHCs and $\{Zn[N(SiMe_3)_2]_2\}$.[31,34]

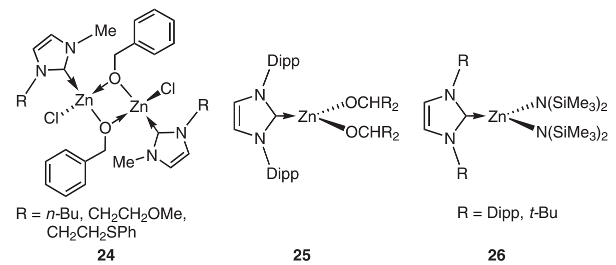
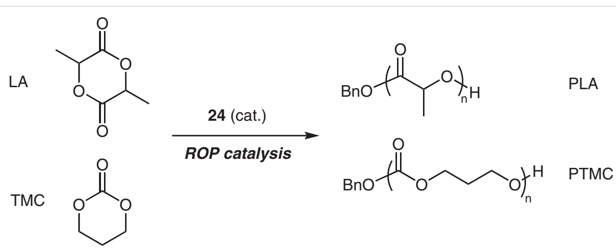


Figure 5 NHC-supported Zn amides and alkoxides **24–26**

4.2 Use in Catalysis

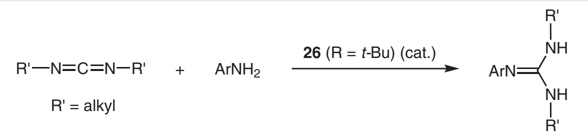
Zn alkoxide complexes **24** were successfully used as effective ROP catalysts of cyclic esters/carbonates such as lactide (LA) and trimethylene carbonate (TMC) (Scheme 16).[16,17] Such catalysis, which proceeds *via* initial insertion/ring-opening of the Zn–OBn bond and subsequent chain growth, allows the production of chain-length-controlled and narrow dispersity polyesters/polycarbonates.



Scheme 16 Cyclic esters/carbonates polymerization catalysis mediated by **24**

At best, with a 1000:10:1 initial ratio of LA/BnOH/**24**, catalysts of type **24** polymerized LA (920 equiv) within 2 hours at room temperature to afford well-defined PLA (M_n = 12500 g·mol^{-1}, $Đ$ = 1.09) (Scheme 16).[16] The ROP of TMC mediated by **24** (with a 5000:50:1 TMC/BnOH/**24** initial ratio) is also effective and controlled, with the consumption of 3750 equivalents of TMC within 6.5 hours at room temperature to afford monodisperse PTMC of expected chain length (M_n = 34800 g·mol^{-1}, $Đ$ = 1.28) (Scheme 16).[17] All experimental data suggest the non-involvement of the NHC ligand in the ROP process, in line with a rather stable Zn–NHC bond under protic conditions (excess BnOH). Interestingly, in the presence of a large excess of MeOH, species **24** also mediates the controlled degradation of PLA to methyl lactate at room temperature.[16]

Apart from polymerization catalysis, it should also be noted that the (I*t*Bu)Zn amide **26** efficiently catalyzes C–N bond formation *via* addition of amines $ArNH_2$ to RN=C=NR to afford the corresponding guanidines $ArN{=}C(NHR)_2$ (Scheme 17).[34]



Scheme 17 Formation of guanidines catalyzed by adduct **26**

5 NHC-Supported Zinc Dihalide Species

5.1 Synthesis

NHC adducts of ZnX_2 (X = Cl, Br, I) of the type $[(NHC)ZnX_2]_2$ and $[(NHC)Zn(X)_2(L)]$ (L = donor solvent) constitute molecularly well-defined ZnX_2 sources that are potentially useful given the importance of ZnX_2-based reagents/catalysts for various organic transformations. Besides robustness and stability, such NHC-stabilized species are also readily accessible upon combining a 1:1 mixture of free NHC and ZnX_2, and are typically isolated in high yields as $[(NHC)ZnX_2]_2$ dimers or, in the presence of a Lewis base, as mononuclear $[(NHC)Zn(X)_2(L)]$ species. Since 2013, several $[(NHC)ZnX_2]_2$ (**27**) and $[(NHC)Zn(X)_2(L)]$ (**28**) species (Figure 6) have been prepared and, in most cases, further exploited in catalysis (*vide infra*).[26,35,36] A cyclic (alkyl)(amino)carbene (CAAC) adduct of $ZnCl_2$, $[(CAAC)ZnCl_2]$ (**29**, Figure 6), was also prepared from free CAAC and $ZnCl_2$.[37]

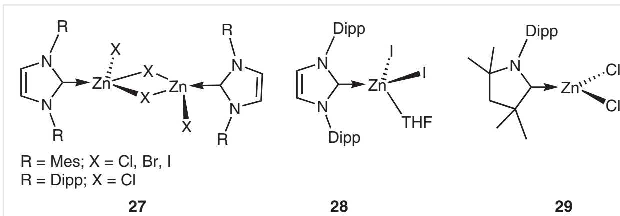
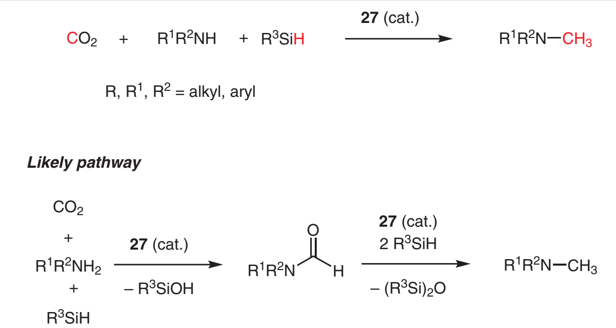


Figure 6 Recently structurally characterized NHC-supported Zn dihalides

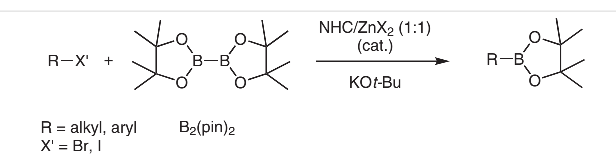
5.2 Use in Catalysis

Adduct **27** (R = Dipp, X = Cl) (Figure 6) was found to catalyze the methylation of various secondary amines in the presence of CO_2 and hydrosilanes to afford the corresponding methylamine products in high yield (Scheme 18).[36] Such a reaction is particularly attractive since it uses CO_2, a cheap and renewable C1 source, and an affordable metal such as zinc for the catalytic formation of C–N bonds through reduction/functionalization chemistry. At best, an *N*-methylaniline/$PhSiH_3$ mixture under CO_2 (1 bar) in the presence of **27** (2 mol%) led to a 95% conversion to MeNRR′ after 20 hours at 100 °C. The use of simple ZnX_2 (X = Cl, Br) salts as catalysts dramatically lowered the reaction efficiency, highlighting the key role of adduct formation for an effective catalytic process. Various control experiments, including the absence of reaction between CO_2 and $PhSiH_3$ in the presence of adduct **27**, led to the conclusion that the reaction proceeds *via* the formation of a formyl intermediate, which was subsequently hydrosilylated to the methylamine product (Scheme 18).



Scheme 18 Catalytic methylation of amines with CO_2 catalyzed by adduct **27**

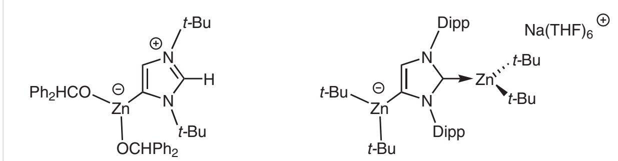
Various adducts of the type $(NHC)ZnX_2$ (X = Cl, Br), generated *in situ* from free NHC and ZnX_2, were recently reported to catalyze the cross-coupling of alkoxy diboron reagent $B_2(pin)_2$ (pin = pinacolate) with alkyl and aryl halides in the presence of KO-*t*-Bu (Scheme 19).[38,39] Such catalysis, which proceeds at room temperature, is most efficient with IMes as the NHC source (Scheme 19). At best, in the presence of $B_2(pin)_2$, KO-*t*-Bu, and an IMes/$ZnCl_2$ mixture (1:1, 15 mol%) as catalyst, a 91% conversion of CyBr to CyBpin was observed within 1 hour at room temperature. With aromatic halides as substrates, the borylation proceeds slower, but may still be performed at room temperature (6–12 h, up to 85% conversion into ArBpin). Although the mechanism of the present Zn-catalyzed borylation remains unclear, experimental data suggest a radical process.



Scheme 19 Catalytic borylation of alkyl/aryl halides with an alkoxy diboron reagent mediated by an NHC–Zn species

6 Other NHC-Stabilized Zn Species

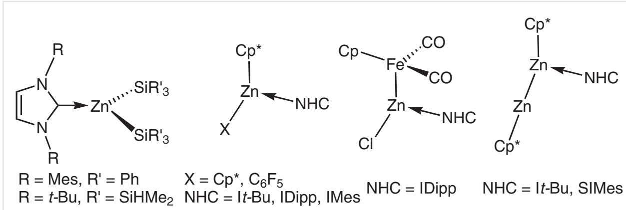
The *a*NHC adduct [(*aIt*Bu)Zn($OCHPh_2$)] (**30**) was serendipitously prepared upon combining sterically bulky carbene I*t*Bu with *in situ* generated '$Zn(OCHPh_2)_2$', a reaction outcome presumably driven by steric relief.[33] The reaction of the sterically congested adduct (IDipp)Zn(t-Bu$)_2$ with [(TMEDA)NaZn(TMP)(*t*-Bu$)_2$] (TMP = 2,2,6,6-tetramethylpiperidide) leads to the formation of dinuclear NHC–Zn anion **31** (as a Na(THF$)_6^+$ salt) *via* a deprotonation/coordination sequence at the NHC C4/C5 position of [(IDipp)Zn(*t*-Bu$)_2$] (Figure 7).[14]



30 31

Figure 7 *a*NHC-supported Zn(II) adducts **30** and **31**

Several other Zn–NHC species have been structurally characterized (Figure 8), including a couple of Zn disilyls of the type [(NHC)Zn($SiR_3)_2$] (**32**),[40,41] NHC adducts of Cp*-based Zn(II) species **33**,[28,42] a Zn–Fe heterobimetallic species **34**,[43] and the only known NHC–Zn(I) species **35**, arising from NHC coordination to Cp*Zn–ZnCp*.[33]



32 33 34 35

Figure 8 Other NHC-supported Zn species **32**–**35**

7 Conclusion

In view of the recent developments highlighted above, NHC-stabilized organozinc species undoubtedly hold promise for use in homogeneous catalysis and fundamental organometallic chemistry. Through NHC coordination, access to more robust and novel Zn-based electrophilic species, including unprecedented Zn hydride and two-coordinate Zn–alkyl cations, was recently achieved. This has thus far led to developments in Zn-catalyzed reduction reactions (essentially hydrosilylation and hydrogenation thus far) of various unsaturated small molecules (including CO_2). Simple NHC adducts of Zn(II) dihalides also appear as effective catalysts for the reductive amination of CO_2 and borylation of alkyl/aryl halides. Besides incorporating an inexpensive and less toxic metal source, NHC-stabilized Zn species benefit from several attractive features that should promote their further use in the coming years, including (a) ready, straightforward, and high-yield synthesis, (b) significantly enhanced stability leading to an optimal reactivity/stability balance, and (c) the great structural variety of available NHCs, resulting in making the steric and electronic properties of the derived Zn(II) complexes easily tunable for novel reactivity/applications.

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References

(1) Arduengo, A. J. III. *Acc. Chem. Res.* **1999**, *32*, 913.
(2) Bourissou, D.; Guerret, O.; Gabbaï, F. P.; Bertrand, G. *Chem. Rev.* **2000**, *100*, 39.
(3) Herrmann, W. A.; Köcher, C. *Angew. Chem. Int. Ed.* **1997**, *36*, 2162.
(4) Arnold, P. L.; Casely, I. J. *Chem. Rev.* **2009**, *109*, 3599.
(5) Bellemin-Laponnaz, S.; Dagorne, S. *Chem. Rev.* **2014**, *114*, 8747.
(6) Fliedel, C.; Schnee, G.; Avilés, T.; Dagorne, S. *Coord. Chem. Rev.* **2014**, *275*, 63.
(7) Nolan, S. P. *N-Heterocyclic Carbenes in Synthesis*; Wiley-VCH: Weinheim, **2006**.
(8) Díez-González, S. *N-Heterocyclic Carbenes: From Laboratory Curiosities to Efficient Synthetic Tools, RSC Catalysis Series*; RSC Publishing: Cambridge, **2011**.
(9) Hirose, T.; Kodama, K. In *Comprehensive Organic Synthesis*, 2nd ed., Vol. 1; Knochel, P.; Molander, G. A., Eds.; Elsevier: Amsterdam, **2014**, 204–266.
(10) Arduengo, A. J. III.; Rasika Dias, H. V.; Davidson, F.; Harlow, R. L. *J. Organomet. Chem.* **1993**, *462*, 13.
(11) Budagumpi, S.; Endud, S. *Organometallics* **2013**, *32*, 1537.
(12) Schnee, G.; Fliedel, C.; Avilés, T.; Dagorne, S. *Eur. J. Inorg. Chem.* **2013**, 3699.
(13) Waters, J. B.; Turbervill, R. S. P.; Goicochea, J. M. *Organometallics* **2013**, *32*, 5190.
(14) Armstrong, D. R.; Baillie, S. E.; Blair, V. L.; Chabloz, N. G.; Diez, J.; Garcia-Alvarez, J.; Kennedy, A. R.; Robertson, S. D.; Hevia, E. *Chem. Sci.* **2013**, *4*, 4259.
(15) Naktode, K.; Anga, S.; Kottalanka, R. K.; Nayek, H. P.; Panda, T. K. *J. Coord. Chem.* **2014**, *67*, 236.
(16) Fliedel, C.; Vila-Viçosa, D.; Calhorda, M. J.; Dagorne, S.; Avilés, T. *ChemCatChem* **2014**, *6*, 1357.
(17) Fliedel, C.; Mameri, S.; Dagorne, S.; Avilés, T. *Appl. Organomet. Chem.* **2014**, *28*, 504.
(18) Collins, L. R.; Moffat, L. A.; Mahon, M. F.; Jones, M. D.; Whittlesey, M. K. *Polyhedron* **2016**, *103*, 121.
(19) Zheng, X.-X.; Zhang, C.; Wang, Z.-X. *J. Organomet. Chem.* **2015**, *783*, 105.
(20) Xu, S.; Everett, W. C.; Ellern, A.; Windus, T. L.; Sadow, A. *Dalton Trans.* **2014**, 14368.
(21) Specklin, D.; Fliedel, C.; Gourlaouen, C.; Bruyere, J.-C.; Avilés, T.; Boudon, C.; Ruhlmann, L.; Dagorne, S. *Chem. Eur. J.* **2017**, *23*, 5509.
(22) Specklin, D.; Hild, F.; Fliedel, C.; Gourlaouen, C.; Veiros, L. F.; Dagorne, S. *Chem. Eur. J.* **2017**, *23*, 15908.
(23) Oestreich, M.; Hermeke, J.; Mohr, J. *Chem. Soc. Rev.* **2015**, *44*, 2202.
(24) Wiegand, A.-K.; Rit, A.; Okuda, J. *Coord. Chem. Rev.* **2016**, *314*, 71.
(25) Rit, A.; Spaniol, T. P.; Maron, L.; Okuda, J. *Angew. Chem. Int. Ed.* **2013**, *52*, 4664.
(26) Rit, A.; Wiegand, A.-K.; Spaniol, T. P.; Okuda, J. *Eur. J. Inorg. Chem.* **2018**, 1114.
(27) Rit, A.; Spaniol, T. P.; Okuda, J. *Chem. Asian J.* **2014**, *9*, 612.
(28) Jochmann, P.; Stephan, D. W. *Angew. Chem. Int. Ed.* **2013**, *52*, 9831.
(29) Rit, A.; Spaniol, T. P.; Maron, L.; Okuda, J. *Organometallics* **2014**, *33*, 2039.
(30) Lummis, P. A.; Momeni, M. R.; Lui, M. W.; McDonald, R.; Ferguson, M. J.; Miskolzie, M.; Brown, A.; Rivard, E. *Angew. Chem. Int. Ed.* **2014**, *53*, 9347.
(31) Roberts, A. J.; Clegg, W.; Kennedy, A. R.; Probert, M. R.; Robertson, S. D.; Hevia, E. *Dalton Trans.* **2015**, *44*, 8169.
(32) Rit, A.; Zannardi, A.; Spaniol, T. P.; Maron, L.; Okuda, J. *Angew. Chem. Int. Ed.* **2014**, *53*, 13273.
(33) Jochmann, P.; Stephan, D. W. *Chem. Eur. J.* **2014**, *50*, 8370.
(34) Baishya, A.; Barman, M. K.; Peddarao, T.; Nembenna, S. *J. Organomet. Chem.* **2014**, *769*, 112.
(35) Al-Rafia, S. M. I.; Lummis, P. A.; Swarnakar, A. K.; Deutsch, K. C.; Ferguson, M. J.; McDonald, R.; Rivard, E. *Aust. J. Chem.* **2013**, *66*, 1235.
(36) Jacquet, O.; Frogneux, X.; Das Neves Gomes, C.; Cantat, T. *Chem. Sci.* **2013**, *4*, 2127.
(37) Singh, A. P.; Samuel, P. P.; Roesky, H. W.; Schwarzer, M. C.; Frenking, G.; Sidhu, N. S.; Dittrich, B. *J. Am. Chem. Soc.* **2013**, *135*, 7324.
(38) Bose, S. K.; Fucke, K.; Liu, L.; Steel, P. G.; Marder, T. B. *Angew. Chem. Int. Ed.* **2014**, *53*, 1799.
(39) Bose, S. K.; Marder, T. B. *Org. Lett.* **2014**, *16*, 4562.
(40) Lemmerz, L. E.; Spaniol, T. P.; Okuda, J. *Z. Anorg. Allg. Chem.* **2016**, *642*, 1269.
(41) Lampland, N. L.; Ellern, A.; Sadow, A. D. *Inorg. Chim. Acta* **2014**, *422*, 134.
(42) Jochmann, P.; Stephan, D. W. *Organometallics* **2013**, *32*, 7503.
(43) Mazzacano, T. J.; Bagherzadeh, S.; Mankad, N. P. *Organometallics* **2013**, *32*, 3986.